

Assignment 1

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Course: UCT702 – IT Workshop (MATLAB)

Title: *Tumor Detection in Medical X-ray Images using Image Enhancement and Segmentation Techniques*

1. INTRODUCTION

The objective of this assignment was to create a MATLAB-based system for tumor detection that can enhance, segment, and assess tumor areas in X-ray images.

Medical X-rays frequently experience issues such as low contrast, significant noise, and indistinct tumor edges, which complicate manual analysis.

This project employs a systematic image processing pipeline that incorporates noise reduction, contrast enhancement, adaptive edge detection, and clustering-based segmentation.

The aim was to produce a distinct binary mask of the tumor and verify its accuracy by using evaluation metrics in comparison to a manually generated ground truth image.

2. METHODOLOGY

2.1 Image Enhancement

To enhance the visibility of tumors, a series of enhancement steps were implemented:

1. **Median Filtering** (3×3–5×5): Eliminated salt-and-pepper noise while maintaining edge integrity.
2. **Gaussian Smoothing** ($\sigma = 1.5$): Diminished high-frequency noise, resulting in smoother intensity transitions.
3. **CLAHE (Contrast Limited Adaptive Histogram Equalization)**: Improved local contrast, thereby enhancing the distinction between tumor and surrounding tissue.

This approach resulted in a clearer and more balanced grayscale image, ideal for segmentation.

2.2 Edge Detection

The Canny edge detector was employed to identify prominent boundaries.

Adaptive thresholding, which relies on image gradient statistics, guaranteed reliable detection across varying lighting and exposure scenarios.

The resulting edge map distinctly highlighted the ribcage and soft-tissue structures, establishing a basis for segmentation.

2.3 Segmentation

Tumor regions were delineated through K-means clustering, which categorized pixels into three intensity-based clusters: background, healthy tissue, and suspected tumor.

The cluster exhibiting the highest mean intensity was recognized as the likely tumor area.

Subsequent processing steps, such as hole filling, morphological closing, and the removal of small objects, enhanced the tumor mask into a unified, continuous region.

Additionally, Fuzzy C-means (FCM) was investigated as an option to enhance edge smoothness for unclear boundaries.

3. EVALUATION METRICS

To evaluate segmentation quality, the detected tumor mask was compared with a manually marked ground truth mask using standard metrics:

Metric	Value	Interpretation
Accuracy	0.576	~58% of pixels were correctly classified overall.
Sensitivity (Recall)	0.229	About 22.9% of tumor pixels were detected — indicating under-segmentation.
Dice Coefficient	0.135	13.5% overlap with the ground truth — low boundary alignment.

The enhancement and edge detection phases were executed successfully; however, the segmentation process encountered difficulties owing to the comparable intensity levels between the tumor and lung tissue.

The low Dice and Sensitivity metrics imply that, although significant structures were accurately identified, smaller tumor areas were overlooked.

Nonetheless, an overall accuracy exceeding 0.5 suggests that the system effectively differentiated the majority of non-tumor regions.

4. DISCUSSIONS AND IMPROVEMENTS

The pipeline effectively showcased a comprehensive and interpretable tumor detection methodology; however, its quantitative performance indicates potential areas for enhancement.

Limitations:

1. The presence of low intensity contrast and overlapping anatomical structures diminished the precision of segmentation.
2. Traditional clustering techniques (**K-means/FCM**) encounter difficulties when dealing with subtle variations in grayscale.

Proposed Enhancements:

1. Integrate texture features (*such as local entropy or variance*) to improve tissue differentiation.
2. Implement adaptive thresholding to achieve more precise boundary extraction.
3. Employ hybrid approaches (**K-means combined with FCM**) or deep learning segmentation techniques (like U-Net) to attain greater accuracy.

Despite these limitations, the implemented approach remains valuable for educational and research purposes, as it is lightweight, explainable, and reproducible on standard MATLAB setups.

5. CONCLUSION

This assignment effectively created a MATLAB-driven tumor detection pipeline that incorporates image filtering, histogram equalization, edge detection, and segmentation.

The pipeline adeptly enhanced and processed X-ray images, offering both visual and quantitative insights into tumor localization.

Although the performance metrics (Accuracy: 0.576, Sensitivity: 0.229, Dice: 0.135) reflect a moderate level of success, the experiment clearly illustrates the potential of traditional image processing as a basis for more sophisticated medical image analysis methods.

By fine-tuning parameters and integrating hybrid or machine learning-based improvements, this strategy can be expanded for enhanced diagnostic applications.

6. VISUAL RESULTS

Original X-ray Image:



Ground Truth Image:



Output Result:

